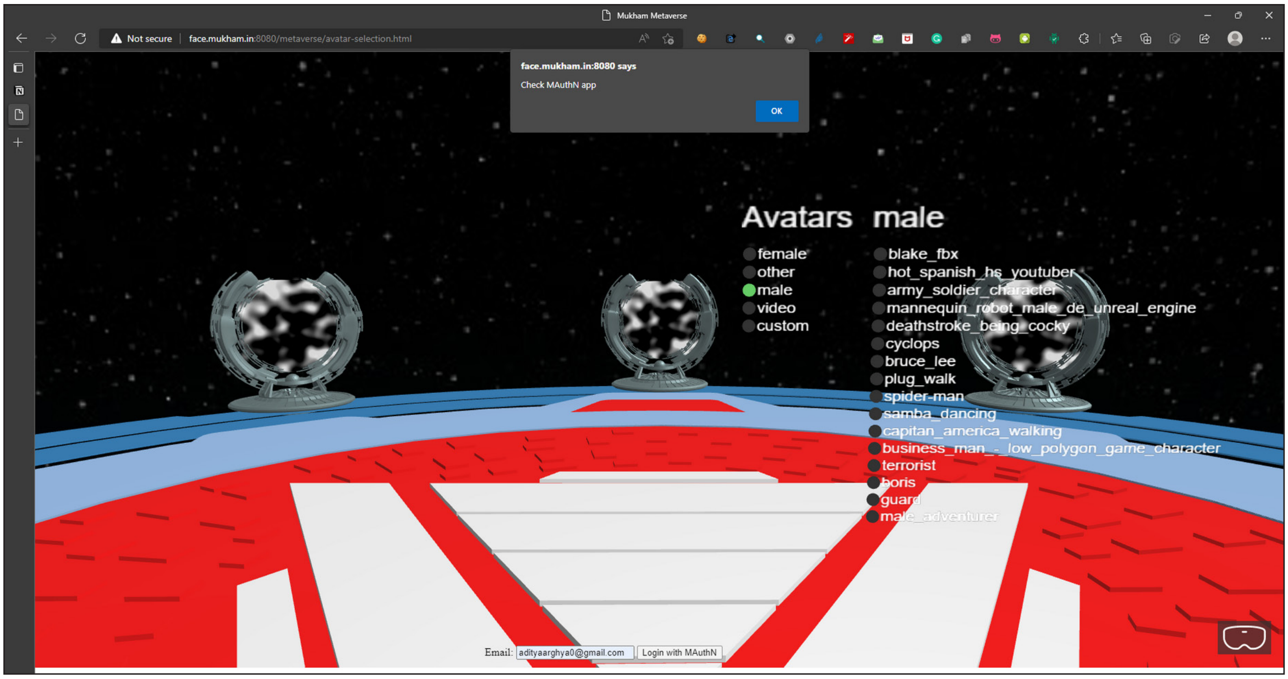

Figure 3: Authentication request with SDK

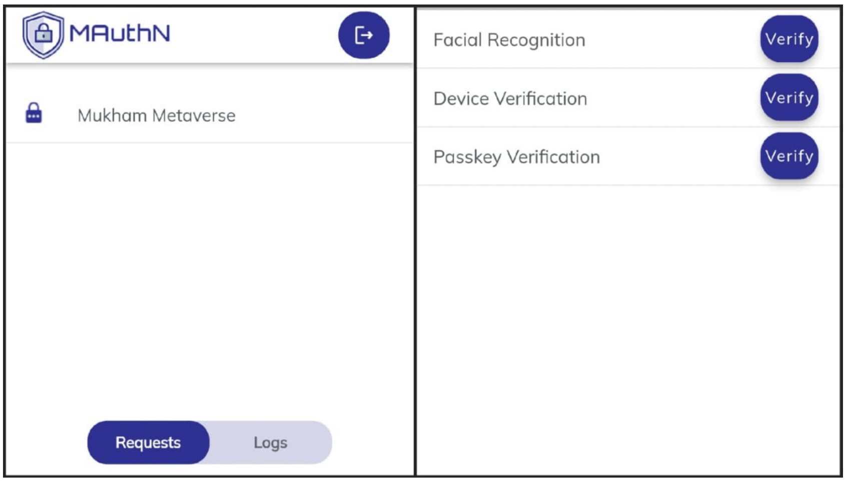

Figure 4: Request on mobile app (left), tri-layer authentication (right)

male Meta Horizon Venues avatars. Similar claims were made by another 21-year old woman. There are numerous instances of cyberstalking, harassment, and misconduct by users. Strongly identifying such users is important for solving such cybercrimes, which may lead to long-term trauma for the victims.

Passwords are today the most used authentication method for gaining access to numerous online services and resources. However, these have several security flaws, such as keylogging, phishing, vishing, and social engineering, making them unsuitable for safeguarding crucial digital assets. The Metaverse relies more on the usage of virtual reality headsets than on conventional keyboards and mice, so passwords are not particularly convenient. From a Metaverse perspective, passwordless, cryptographic, and biometric-based authentication is far more suitable.

MetaKey: The solution

MetaKey is a unique solution to provide a high level of security in the Metaverse, and can be used to securely identify and authenticate a user in this virtual world. In the Metaverse, physical security keys are needed for authenticating a user. This can be done with the help of FIDO Alliance FIDO2 specifications, which give a set of protocols (namely WebAuthn and CTAP) for strongly authenticating the user. They provide Authentication Assertion Level 3 (AAL3), which, according to the US based National Institute of Standards and Technology (NIST), is the strongest form of authentication a system can have. This mitigates the chances of lost credentials, or account takeovers, as the user needs to physically connect a smart card or a security key to his or her device (the computer, smartphone, or VR headset) to log in successfully.

However, strongly authenticating a user does not completely mitigate all the security risks. (What if the smart card of a user is stolen?) Thus, the solution also suggests using facial recognition to identify the user. Facial